



Customer Shopping Behavior Analysis

3,900 transactions · 18 features · Strategic insights across demographics, spending, and product preferences

Dataset at a Glance

3,900

Transactions

37

Missing Values

Review Rating only

18

Features

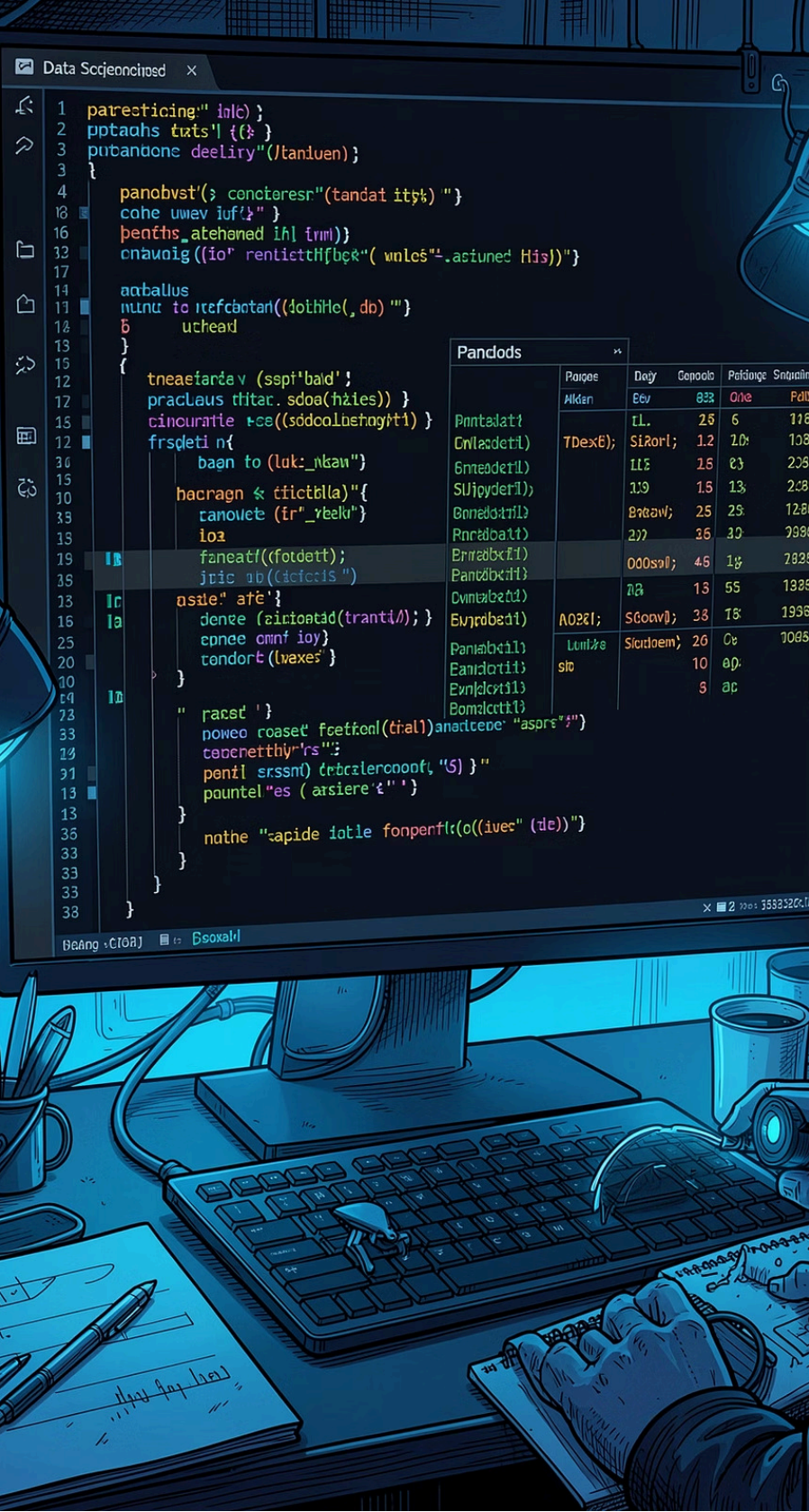
50

Locations



Key Feature Groups

- **Demographics:** Age, Gender, Location, Subscription
- **Purchases:** Item, Category, Amount, Season
- **Behavior:** Discount, Promo Code, Frequency, Rating



Data Preparation in Python

01

Load & Explore

Imported via pandas; `df.info()` and `.describe()`

02

Clean

Imputed Review Rating by category median; standardized to snake_case

03

Engineer

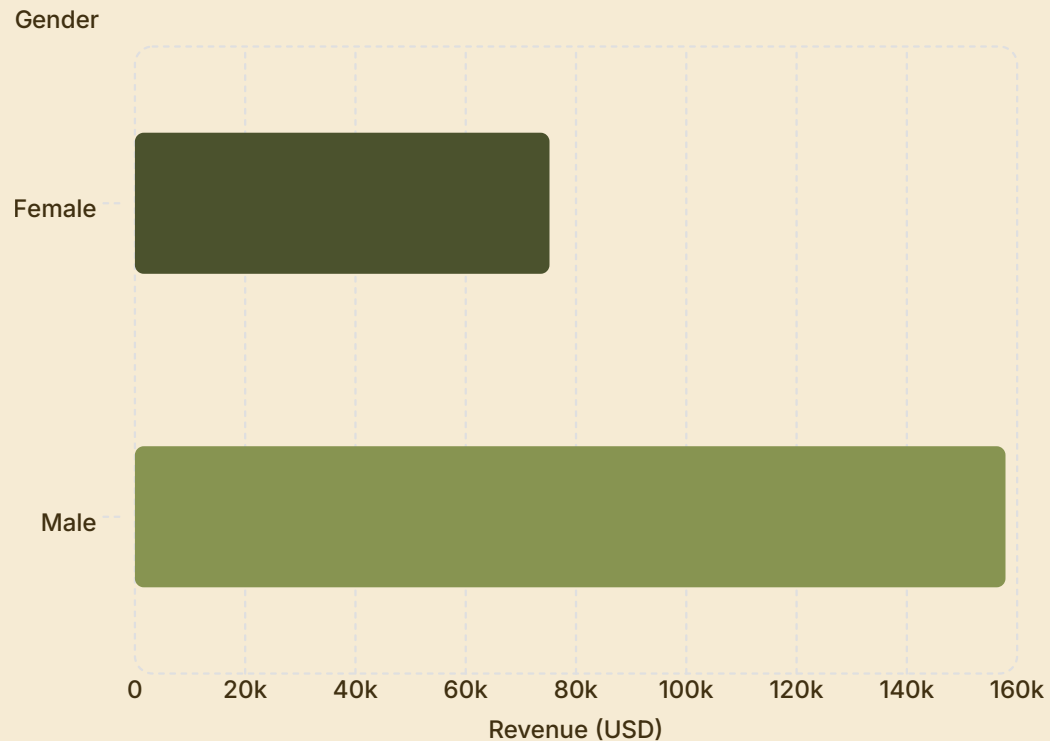
Built `age_group` and `purchase_frequency_days`; dropped redundant `promo_code_used`

04

Load to DB

Connected to PostgreSQL for SQL analysis

Revenue by Gender

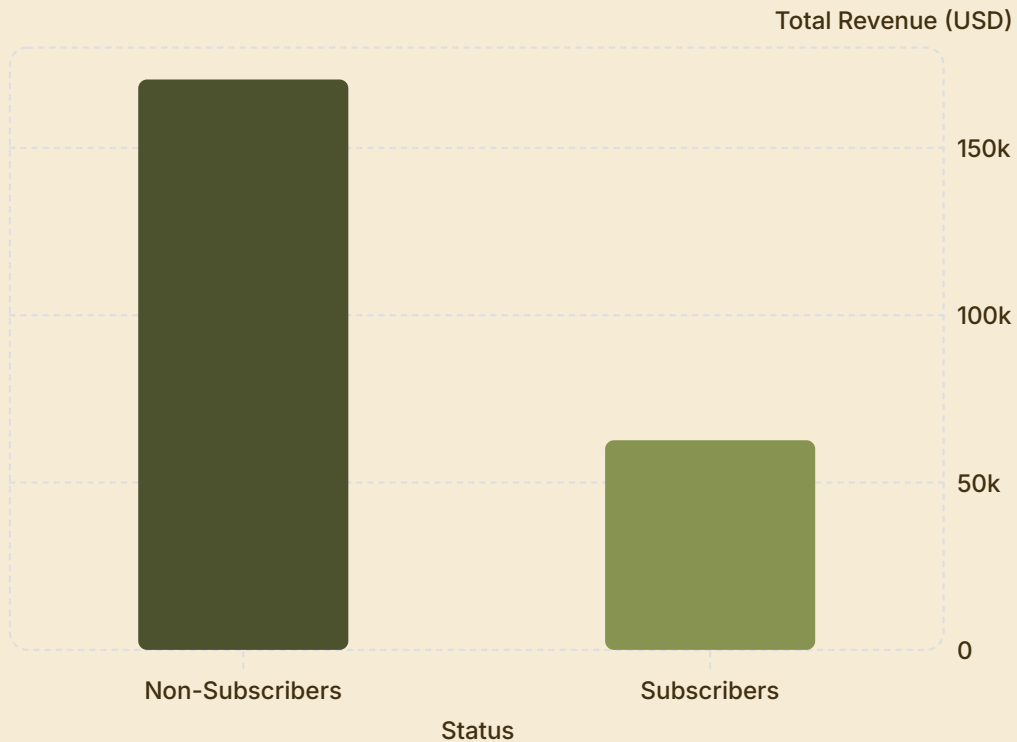


Male customers drive 2× revenue

Male: **\$157,890** · Female: **\$75,191**

Male purchases also outnumber female 2,652 vs. 1,248 — a significant behavioral gap to address.

Subscribers vs. Non-Subscribers

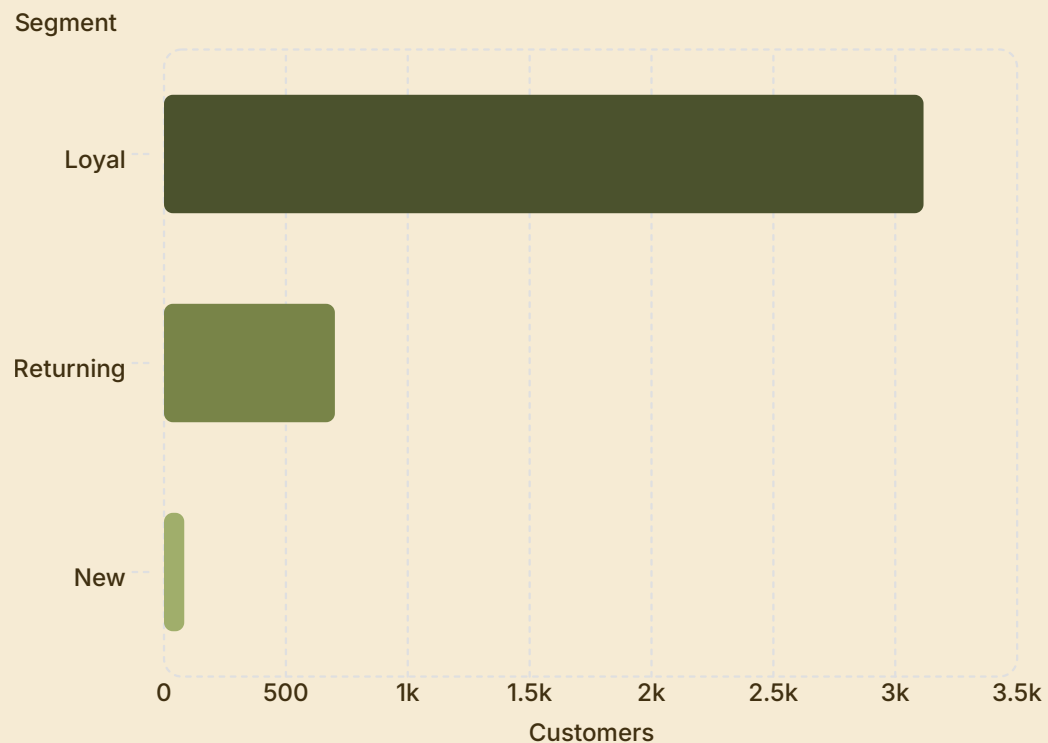


73% of customers are not subscribed

Non-subscribers generate **\$170K** vs. subscribers' **\$63K**

Average spend is nearly identical (~\$59.50), signaling a major conversion opportunity.

Customer Segmentation



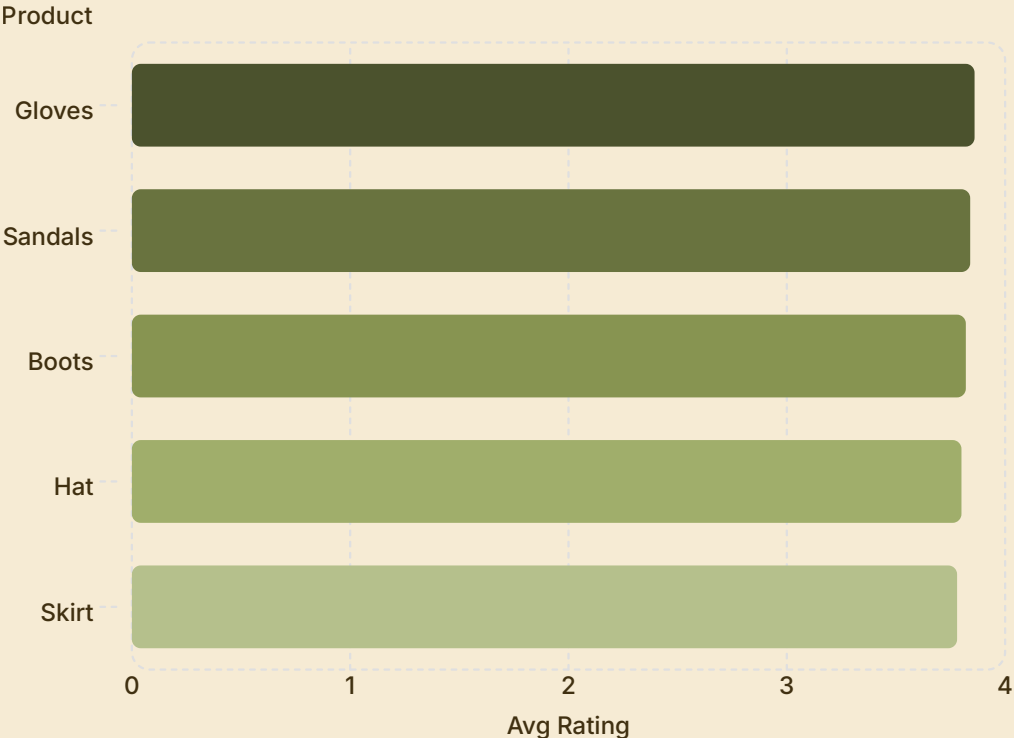
Loyal customers dominate

3,116 Loyal · 701 Returning · 83 New

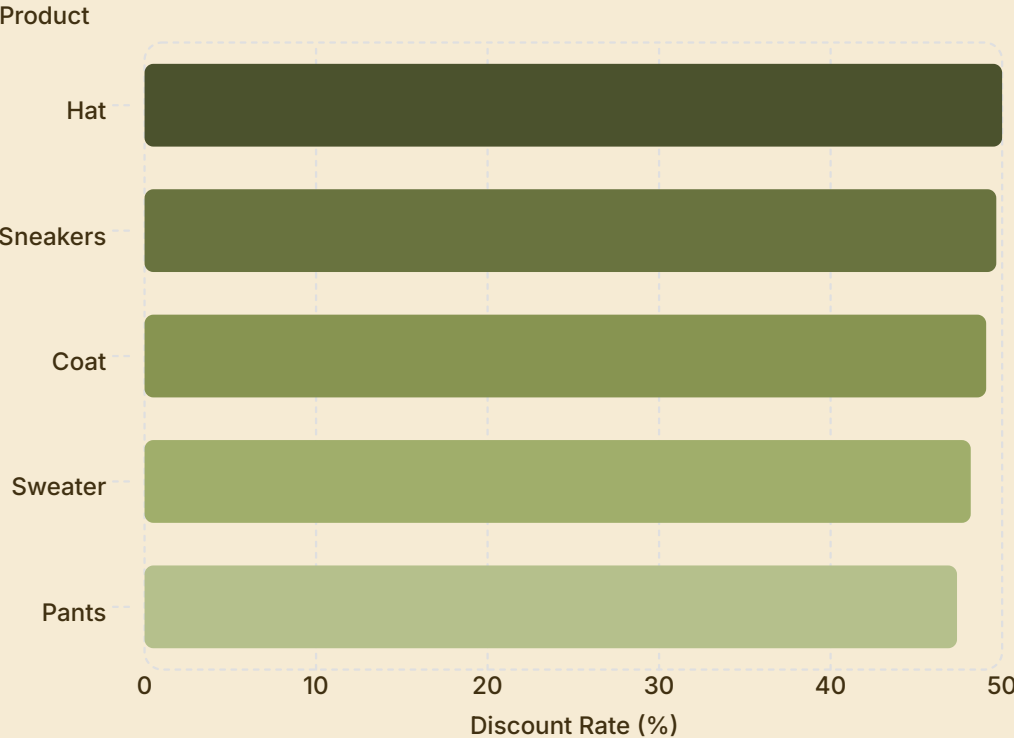
Of 958 repeat buyers (>5 purchases), only ~30% subscribe — a key retention lever.

Top Products by Rating & Discount Rate

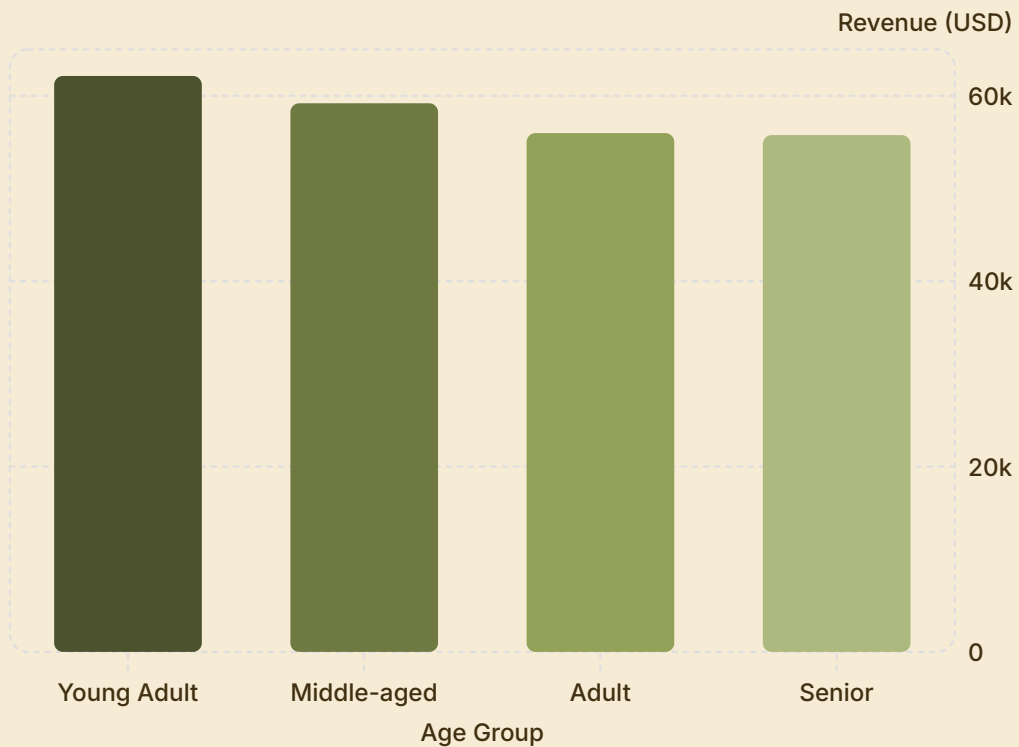
Highest Rated



Most Discounted



Revenue by Age Group



Young Adults lead revenue

Young Adult: **\$62,143** — highest across all age groups

Revenue declines gradually with age, but all groups contribute meaningfully.

Power BI Dashboard



Key Metrics

Visualizes revenue by category, age group, subscription status, shipping type, and gender.

3.9K
Customers

\$59.76
Avg Purchase

3.75
Avg Rating

Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers toward "Loyal" segment



Review Discounts

Balance sales lift with margin control



Targeted Marketing

Focus on high-revenue age groups & express-shipping users

